

Analytic Combinatorics of Unlabeled Objects

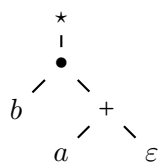
Set of exercises 2

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1. Warm up: regular expressions*

Consider the combinatorial class $\mathcal{R}$ of regular expressions as unary-binary trees:

- leaves are decorated by the letters “ a ”, “ b ” or the empty word “ ε ”.
- internal binary nodes are decorated by the operators union “ $+$ ” or concatenation “ $\bullet$ ”,
- internal unary-nodes are decorated by the Kleene-star “ $\star$ ”.



$$(b \cdot (a + \varepsilon))^*$$

Give a combinatorial specification. Determine the ordinary generating function supposing that the size is the total number of nodes.

2. Multivariate Generating Functions

A) Integer compositions*

We recall that integer compositions are specified by

$$\mathcal{C} = \text{Seq}(\{1, 2, 3, \dots\}) \simeq \text{Seq}(\text{Seq}_{\geq 1}(\mathcal{Z})).$$

During class, we used OGFs to prove that the average number of terms in a composition of n is $\sim n/2$.

1. What if we restricted the compositions to have an even number of terms? Find the total number of compositions of n into an even number of terms.
2. Let us consider now the number of even terms in a general composition of n . Show that the average number of even terms is $\sim n/6$.
3. Now we want to fix the number m of even terms. Find the generating function.

B) Binary trees

We consider the binary trees

$$\mathcal{B} = \mathcal{E} + \mathcal{Z} \times \mathcal{B} \times \mathcal{B},$$

where the node $\mathcal{E}$ is the empty-node, while $\mathcal{Z}$ is an atom of weight 1.

1. Give a specification for the binary trees $\mathcal{B}^\circ$ marking the length of the left-most branch.
2. Solve the equation for $B(z, u)$ and show $\partial_u B(z, 1) = z(B(z))^3$.
3. Let $A(z) = zB(z)$. Apply the Lagrange Inversion Formula to compute $[z^n](A(z))^3$.
4. Deduce that the average length of the left-most branch¹ tends to 3 as $n \rightarrow \infty$.
5. Generalize. Find the exact number of binary trees of size n with left-most branch of length m .

3. Rational asymptotics**Find the generating function***

Find the ordinary generating function of the following regular languages:

1. The class of words on $\{a, b\}$ such that, between any two b 's there is at least one a .
2. The class of words on $\{a, b, c\}$ such that

$$\#a + \#b + 2\#c \equiv 0 \pmod{3}.$$

3.1. Schur's Theorem

Prove the following theorem.

Schur's Theorem. If c_n represents the number of representations of n as a non-negative integer combination of $a_1, \dots, a_M$, these being a set of positive integers with $\gcd(a_1, \dots, a_M) = 1$, then

$$c_n \sim \frac{n^{M-1}}{(M-1)! a_1 \dots a_M}.$$

Hint. What $\gcd(1 - x^{a_1}, 1 - x^{a_2}, \dots, 1 - x^{a_M})$?

¹We will later see in class that the average-depth of a random binary tree is rather $\Theta(\sqrt{n})$.